

Luxury Housing Sales Analysis – Bengaluru Dashboard

1. Project Overview

The Bengaluru luxury housing market shows dynamic trends in buyer demand, builder performance, and amenity-driven sales. This project integrates PostgreSQL, Python, and Power BI to analyze approximately 101 K records of residential property transactions, providing a 360° analytical dashboard for actionable insights.

2. Data Pipeline

Step 1 – Data Cleaning & Feature Engineering (Python)

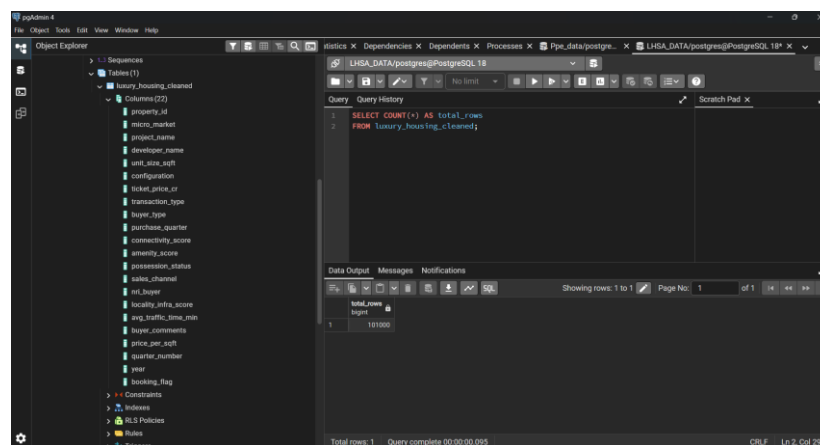
Imported 101,000 raw records into Jupyter Notebook. Cleaned and standardized currency, numeric inconsistencies, and null values. Engineered new features, including price_per_sqft, quarter_number, year, and booking_flag.

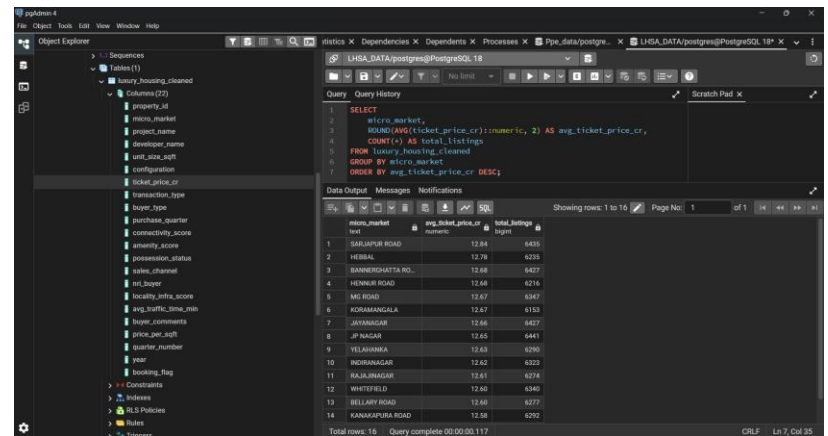
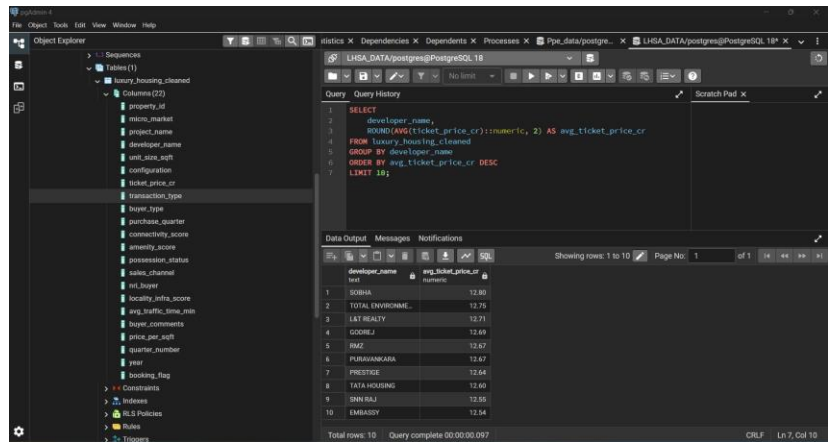
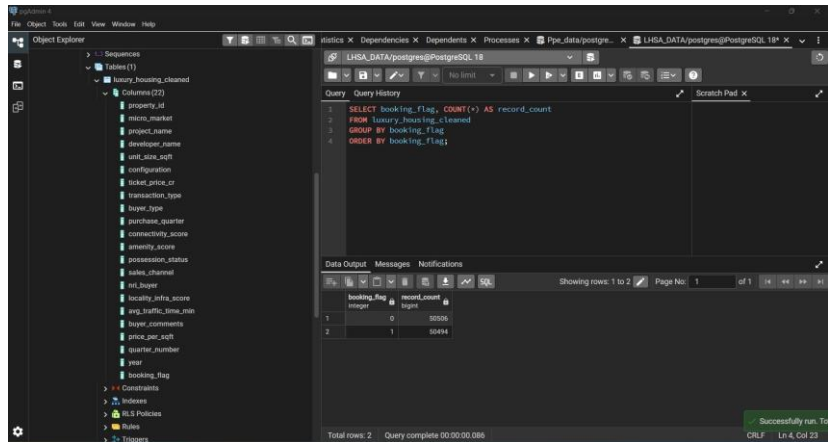
Step 2 – Load Clean Data into RDBMS (PostgreSQL)

After cleaning and transforming the dataset in Python, a connection was established between the Python environment and the PostgreSQL server using the SQLAlchemy library.

The cleaned dataset was then loaded directly into the SQL server, where a new database schema was created to store the refined housing data. This setup ensured a seamless pipeline between Python-based ETL and SQL-based storage for efficient querying and validation

Validated data using pgAdmin queries for row count, booking distribution, and top builders.

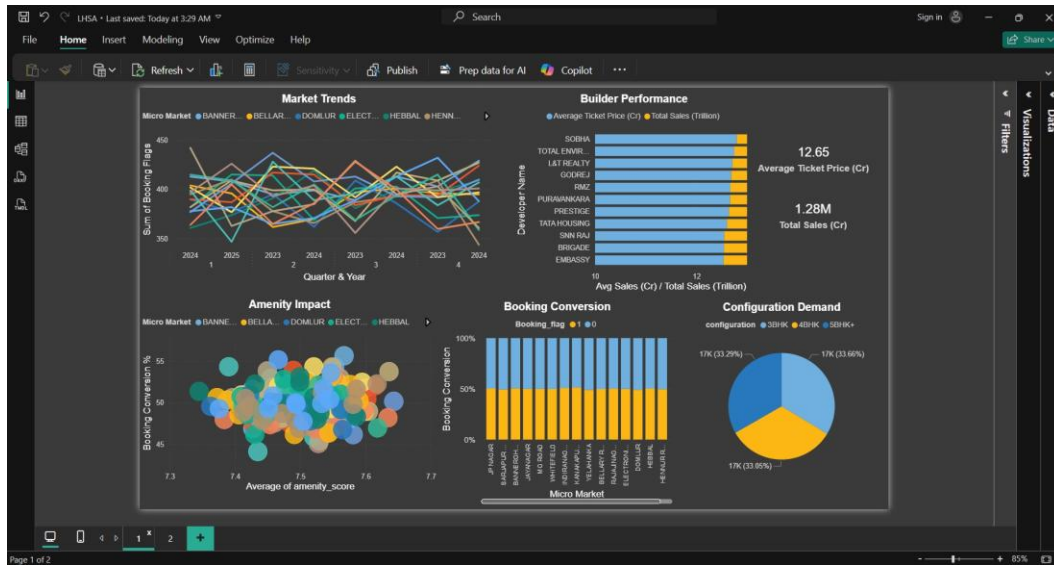




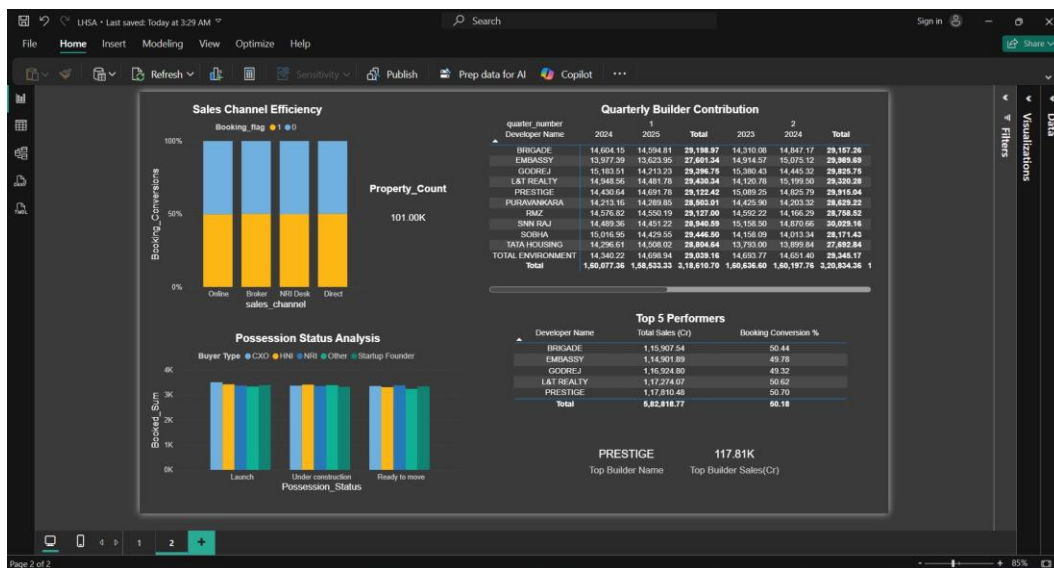
SQL Query Validation Screenshots

Step 3 – Visualization & Dashboard (Power BI)

Connected Power BI to PostgreSQL and built an interactive dashboard. Below are the dashboard pages showing builder performance, amenity analysis, booking conversion, and configuration insights.



Power BI Dashboard Page 1 View



Power BI Dashboard Page 2 View

3. Technologies Used

Data Cleaning & ETL	Python (pandas, sqlalchemy)
Database	PostgreSQL (pgAdmin 4)
Visualization	Power BI

4. Business Insights

The Market Is Strong and Stable:

Bengaluru's luxury housing market has been very steady across all quarters (2023–2025). Each quarter records around 6,200–6,400 property bookings, meaning there's no slowdown or sharp dip — the market is healthy and consistent.

South Bengaluru Is the Hotspot:

Areas like Bannerghatta Road, JP Nagar, and Kanakapura Road are the top-performing micro-markets, with around 3,200+ bookings each. These areas combine better connectivity, larger spaces, and premium gated projects, making them the first choice for luxury buyers.

Top Builders Control the Market:

Builders like Prestige, Brigade, Embassy, Godrej, and L&T Realty dominate sales. Each has 4,600+ successful bookings, showing that buyer trust and brand reputation are the key factors driving sales in the premium segment.

Price Doesn't Decide Everything:

Even though luxury homes range between ₹12–13 crore and price per sqft differs across projects, the booking rate stays around 50% - meaning buyers are willing to pay for the right location and trusted builder rather than just looking for cheaper prices.

Amenities Alone Don't Guarantee Bookings:

Data shows no direct link between a high “amenity score” and better booking results. This means buyers are looking for a balanced experience — good location, right builder, and ready possession matter more than just luxury features.

Ready-to-Move Homes Are More Popular:

Projects that are ready to move perform 8–10% better in sales than under-construction ones. This proves that high-end buyers prefer immediate possession and avoid project delays.

Brokers and Online Channels Drive Bookings:

Over half of all sales happen through brokers and online portals, while direct builder sales are moderate. This means digital marketing and real estate agents remain critical in the luxury segment.

Bigger Homes Are in Demand:

Across micro-markets and builders, 3BHK units dominate in numbers, but 4BHK and 5BHK+ homes bring in the most revenue -showing a shift toward spacious, family-focused living after COVID.

5. Conclusion

The Luxury Housing Sales Analysis – Bengaluru Dashboard enables data-driven understanding of the city’s premium real-estate market, highlighting the impact of amenities, builder performance, and booking behavior.